WARNING

KEEP OUT OF REACH OF CHILDREN

BACSTAR ™

BIO-FUNGICIDE

Active Ingredient:
BACSTAR contains 250g/kg dried *Bacillus subtilis var. amyloliquefaciens strain* D747 and contains not less than 5.0 X 10^{10} cfu/g in the form of a water dispersible granule.

For the control and suppression of powdery mildew, Botrytis and sour rot on Grapes, fireblight on Pipfruit, Botrytis on Berryfruit, Kiwifruit, Persimmons, and Onions, and various diseases on Avocados, Citrus, Lettuce, Turf and Kiwifruit

GROUP	44	FUNGICIDE

NET CONTENTS: 2.5 kg

CERTIS
The Biopesticide Company

Registered pursuant to the ACVM Act 1997, No. P008310
See www.foodsafety.govt.nz for registration conditions
Approved pursuant to the HSNO Act 1996, Approval Code HSR100430
Biogro certified organic 4429.

UPL New Zealand Limited
PO Box 51584
Pakuranga 2140
Auckland
Phone: 0800 100 325

In a transport emergency dial 111, Police or Fire Brigade.
For specialist advise in an emergency only, call 0800 CHEMCALL **(0800 243 622)**

UN-Free

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NEW ZEALAND
18/07/2025

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PLEASE READ THE LABEL COMPLETELY BEFORE USE
This product MUST NOT be used for any purpose or in any manner contrary to the controls and conditions of this registration.

HSNO Classifications: 6.5B, 9.1D
WARNING
May cause an allergic skin reaction.
Designed for biocidal action.

PRECAUTIONS
Store in original container, tightly closed, away from foodstuffs and under lock and key. Avoid contact with seeds, fertilisers and other pesticides.
Avoid contact with skin, eyes and inhalation of spray mist.
When spraying, wear overalls and gloves.
DO NOT eat, drink or smoke while using.
Wash hands and face before eating, drinking or smoking and after work.
Wash protective clothing after use.
Apply with well-maintained and calibrated spray equipment.
DO NOT carry on a passenger service vehicle.
Application, storage and record keeping should be as described in the Management of Agrichemicals NZS8409.
Avoid contamination of any water supply with chemical or empty container.

Container disposal: Ensure the container is empty. Triple rinse empty container and add rinsate to the spray tank. Otherwise crush and bury in a suitable landfill. DO NOT reuse this container for any other purpose.

Product disposal: Dispose of this product only by using according to the label, or at an approved landfill or other approved facility.

FIRST AID
If swallowed DO NOT induce vomiting. For advice, contact the National poisons Centre 0800 POISON (0800 764 766) or a doctor immediately. If skin or hair contact occurs, remove contaminated clothing and flush skin and hair with running water. DO NOT scrub the skin. If splashed in eyes, wash out immediately with water. If inhaled move the victim to fresh air immediately. Begin artificial respiration if breathing has stopped.

SPILLAGE
Wear suitable protective clothing including face shield, impervious gloves and boots. Prevent the product or spilled material from entering water bodies. Absorb liquid spills with inert material such as earth or sand and place in waste containers. Wash area with water and absorb with further inert material. Dispose of safely.

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GENERAL INFORMATION:

BACSTAR™ has been developed from naturally occurring strains of *Bacillus subtilis*. It is a broad-spectrum preventative fungicide recommended for the control and suppression of powdery mildew, *Botrytis* and sour rot on Grapes, fireblight on Pipfruit, *Botrytis* on Berryfruit, Kiwifruit, Persimmons, and Onions, and various diseases on Avocados, Citrus, Lettuce, and Turf during periods of low disease infection pressure. Against fungal diseases, BACSTAR acts by preventing spore germination, germ tube elongation and germ tube penetration.

BACSTAR can be integrated into an overall disease and pest management strategy wherever fungicide use is necessary in an IPM programme.

Thorough coverage is essential for optimum disease control. The addition of an adjuvant is highly recommended particularly on hard to wet surfaces. Use DU-WETT or DU-WETT WEATHERMAX in low water volumes and BOND XTRA or ACTIWETT in high water volumes.

BACSTAR is also an important tool in resistance management strategies. BACSTAR should be applied prior to or in the early stages of disease development.

RESISTANCE MANAGEMENT

BACSTAR contains a microbial fungicide (FRAC Group 44). Resistance to BACSTAR could develop from repeated use. To minimise this risk, use strictly in accordance with the label instructions and Directions for Use. BACSTAR should be used in a program which incorporates effective fungicides with alternate modes of action.

DIRECTIONS FOR USE

Thorough coverage is essential for optimum disease control. To achieve good coverage, use the correct adjuvant, spray pressure, water volume nozzle-size and spacing with the correct ground speed are critical. Sprayers should be calibrated before use. To do this consult spray nozzle catalogues and other industry references for specific information on sprayer calibration or contact a registered 'sprayer calibration' consultant.

BACSTAR is intended for use as either a dilute spray or as a concentrate spray with DU-WETT super spreader technology through conventional ground spray equipment.

Crop	Disease	Remarks
Note: For the control of any disease, use alternative fungicides when disease pressure is high and during the rest of the season prior to harvest.		
Apples, Pears	Fireblight	Make the first application up to 5% blossom and apply 100 - 150 g/100 litres (or 1.0 – 1.5kg/ha) following a streptomycin application to minimise disease resistance to antibiotics. Under severe disease infection periods, repeat applications may be made at 3-7 days interval, using the longer interval after petal fall. The addition of a suitable adjuvant (eg DU-WETT®) is recommended. The addition of a copper product may enhance activity.
Avocados	Anthracnose	Apply 80g/100L in a tank mixture with the lowest rate of a copper-based fungicide, or 100g/100 litres alone over flowering. The addition of a suitable adjuvant (eg DU-WETT®) is recommended.
The use rates for all subsequent crops are 1.0 – 2.0kg /ha or 100 - 200 g/100 litres in sufficient water to ensure complete coverage of plants and flower parts. The lower rates are generally for preventative applications and higher rates for conditions favouring disease infection.		
Citrus	*Botrytis,*	Apply at 50% flowering and repeat at 7-10 days interval if disease conditions persist. The addition of a suitable adjuvant (eg DU-WETT®) is recommended.
	Anthracnose,	Apply at monthly intervals from early flowering to harvest.
	Greasy spot,	For suppression, begin applications at first leafy flush, and repeat at later flushes. Include spray oil in tank mixture. Can also be tank mixed with other fungicides including copper-based ones.
	Melanose	For suppression, begin applications at petal fall and repeat at 14-21 days interval.

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Grapes	Powdery mildew, *Botrytis*, Sour rot	BACSTAR can be used all season particularly in organic crops. Use no more than 2 applications back to back without alternating with other products. Under high pressure in conventional blocks use the best available alternatives at critical timings 80% cap fall and bunch closure. BACSTAR can be used later in the season as it has nil residues and no withholding period restrictions. Two strategically timed applications of BACSTAR are recommended at 4-5 weeks from harvest and again at 1 week from harvest. Use the higher rate and shorter interval when conditions favour rapid disease development. Apply in sufficient water to ensure thorough coverage of all parts of the plants early to mid season and in late season applications, target the bunch line area. The addition of a suitable adjuvant (eg DU-WETT®) is recommended. The addition of a Sulphur product may enhance activity on Powdery mildew.
Kiwifruit	*Botrytis*	Apply when weather conditions are conducive to infection at late flowering to early fruit set and immediately pre-harvest. Under heavy disease pressure, use in a spray program with other fungicides. May give suppression of *Sclerotinia.* The addition of a suitable adjuvant (eg DU-WETT®) is recommended.
Persimmons	*Botrytis*	Apply over flowering and repeat at 7-10 days interval if disease conditions persist.
Strawberries Berryfruit	*Botrytis*	Apply from early flowering and during harvest at 10-14 days interval if disease conditions persist. Apply in a program of alternation with other fungicides. The addition of a suitable adjuvant (eg DU-WETT®) is recommended.
Lettuce	*Sclerotinia minor*	Does not control *S. sclerotiorum.* Apply as a soil drench immediately before or after transplanting or seedling crop emergence. Repeat as necessary before crop coverage of the soil and alternate with another fungicide for greater disease control spectrum. The addition of a suitable adjuvant (eg DU-WETT®) is recommended.
Onions	*Botrytis,* Downy mildew	As part of a program, apply pre-harvest for the moderate control and suppression of diseases. The addition of a suitable adjuvant (eg DU-WETTt®) is recommended.
Ornamentals	*Botrytis*, Powdery mildew	Apply at the first sign of disease and repeat at 7 – 10 days interval. The addition of a suitable adjuvant (eg DU-WETT®) is recommended.
Turf	Anthracnose, Brown patch, Dollar spot, Powdery mildew, Rust	Apply when conditions are conducive to disease development. Use the higher rate for improved control and the lower rate when in tank mixture with a suitable fungicide or in a rotation program with alternative chemistry. Repeat as necessary at 7-10 days interval. The addition of DU-WETT® adjuvant is strongly recommended.

MIXING

Partially fill the spray tank with clean water and start agitation. Add the specified amount for the area to be treated of BACSTAR to the spray tank. Complete filling the tank to the required volume for the area to be treated. Maintain agitation until spraying is complete. Do not allow the spray mixture to stand overnight or for any prolonged periods.

Spray pH must be not less than pH 4.5 and not greater than pH 8.5.

COMPATABILITY:

The addition of copper and sulphur based products may enhance disease control.

Do not combine BACSTAR in the spray tank with pesticides or fertilisers unless prior experience has shown the combination to be compatible. Any tank combinations must also be tested for effectiveness on disease and selectivity to plants before use on commercial areas. (Refer to notice to buyer).

If the use of any tank mixture is desirable, a small volume of the mixture should first be tested in a small measured volume, for physical compatibility at the dilution rates required. If this test is positive, a field test of the mixture should then be carried out on a few plants to observe effectiveness and plant safety.

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It is an offence to use this product on animals.

It is an offence for users of this product to cause residues exceeding the relevant MRL in the Food Notice: Maximum Residue Limits for Agricultural Compounds

STORAGE
Store in a cool dry place avoiding temperatures above 35°C.

SAFETY DATA SHEET
If additional hazard or safety information is required, refer to the Safety Data Sheet. For a copy visit our website at https://www.uplcorp.com/nz

EXCLUSION OF LIABILITY:
Please read these conditions of supply before buying and using.
If these conditions are unacceptable, return at once unopened.
To the extent permitted by law, buyers and/or users of the goods accept that neither the manufacturer, UPL New Zealand Limited as distributor, nor any other distributor have any liability or responsibility whatsoever for any loss, damage or injury whether in contract or tort, whether direct, indirect or consequential howsoever arising in connection with the supply of these goods.

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Batch No. DOM	Barcode

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